

JIWON KIM

PhD Candidate, Neuroscience · University of Pittsburgh | jik123@pitt.edu | <http://ji-woni.github.io/>

EDUCATION

University of Pittsburgh

Jun 2023 – Present

PhD Candidate, Neuroscience · Center for Neuroscience (CNUP), Pittsburgh, PA

- Advisors: Drs. Alberto Vazquez & Sarah Ross | Passed Comprehensive Exam, Dec 2025 | Expected graduation: 2028

Washington and Lee University

Sep 2016 – May 2020

BS, Neuroscience · Lexington, VA | GPA: 3.79

RELEVANT EXPERIENCE & ACCOMPLISHMENTS

PhD Candidate — Vazquez & Ross Labs, University of Pittsburgh

Jun 2023 – Present

- Designed and executed an independent research project from hypothesis to preliminary data, generating results that secured \$70,676 in AHA Predoctoral Fellowship funding (3.66th percentile) from Jan 2026 to Dec 2027
- Investigating cortical interneuron regulation of angiogenesis and neurovascular coupling with direct implications for stroke recovery and neurodegeneration therapeutics
 - Surgeries: craniotomy, cranial window implantation, stereotaxic injections
 - Imaging: 2P *in vivo* imaging, laser speckle contrast imaging, IHC, *in situ* hybridization
 - Photothrombotic stroke model; MATLAB analysis.
- Contributed key experiments such as *in situ* hybridization to a cross-lab collaboration resulting in co-authored publications in *Science* (2025) and *Science Advances* (in press, 2026)
- Mentoring: trained 2 postdoctoral fellows, 1 principal investigator, and 2 undergraduates in survival surgeries and *in vivo* experiments including 2P imaging and whisker stimulation
- Presented thesis work and rotation findings at 3 conferences over 2 years; awarded \$1,400 competitive travel bursary

Research Assistant — Yuan Lab, Boston Children's Hospital / Harvard Medical School

Aug 2020 – May 2022

- Contributed to 4 published papers across 3 simultaneous projects within a 5-person team, collaborating closely with external labs and clinicians to align experimental findings with patient-relevant outcomes in pulmonary vascular disease
 - *In vivo* mouse models: transgenic mouse husbandry, hypoxia chamber maintenance & monitoring, retroorbital and intraperitoneal injections, perfusion
 - Imaging: IF and IHC, iDISCO tissue clearing, cryosectioning, vibratome sectioning, confocal imaging
 - Molecular biology: *in vitro* cell experiments, RNA isolation, genotyping, FACS/cell sorting
- Mentoring: trained an incoming fellow in mouse husbandry, perfusion, and IHC and onboarded a technician for 6 months to ensure project continuity after my departure
- As one of the first members of a newly established lab, built operational infrastructure carrying administrative responsibilities while conducting independent research

LEADERSHIP

President – Young Adult Group, Korean Central Church of Pittsburgh

Apr 2025 – Apr 2026

- Led a 100+ member young adult ministry comprising 4 teams and 8 small groups
- Managed an annual budget of \$15,000 across programming, outreach, and operations
- Directed full-cycle event planning and execution for large-scale community and worship events

Founder — Vision Unification (Nonprofit organization)

May 2019 – present

- Secured grant funding to launch a nonprofit promoting cultural unification among Koreans
 - Center for International Studies Summer Grant (\$5,000) — Washington and Lee University, 2019
 - Goldman Sachs Fund (\$5,514) — Washington and Lee University, 2018
 - Endeavor Foundation Grant (\$4,250) — Washington and Lee University, 2017

PUBLICATIONS

- A.W. Liu, Y.R. Zhang, ..., **J. Kim**, T.D. Sheahan, S.E. Ross, M. Meisel, T.L. Sumpter, and D.H. Kaplan. "Scratching promotes allergic inflammation and host defense via neurogenic mast cell activation." *Science*, 387(6733):eadn9390, 2025.
- H. Kim, Y. Liu, **J. Kim**, et al. "Pericytes contribute to pulmonary vascular remodeling via HIF2 α signaling." *EMBO Reports*, 2024.
- E. Taglauer, E. Wachman, ..., **J. Kim**, ..., K. Yuan. "Acute SARS-CoV-2 Infection in Pregnancy Is Associated with Placental ACE2 Shedding." *The American Journal of Pathology*, 2022.
- T. Klouda*, Y. Hao*, H. Kim, **J. Kim**, et al. "Interferon-alpha Facilitates SARS-CoV-2 Pulmonary Vascular Infection by Inducing ACE2." *Angiogenesis*, 2021.
- T. Klouda, H. Kim, **J. Kim**, G. Visner, K. Yuan. "Precision Cut Lung Slices as an Efficient Tool for Ex Vivo Pulmonary Vessel Studies." *Journal of Visualized Experiments*, 2021.